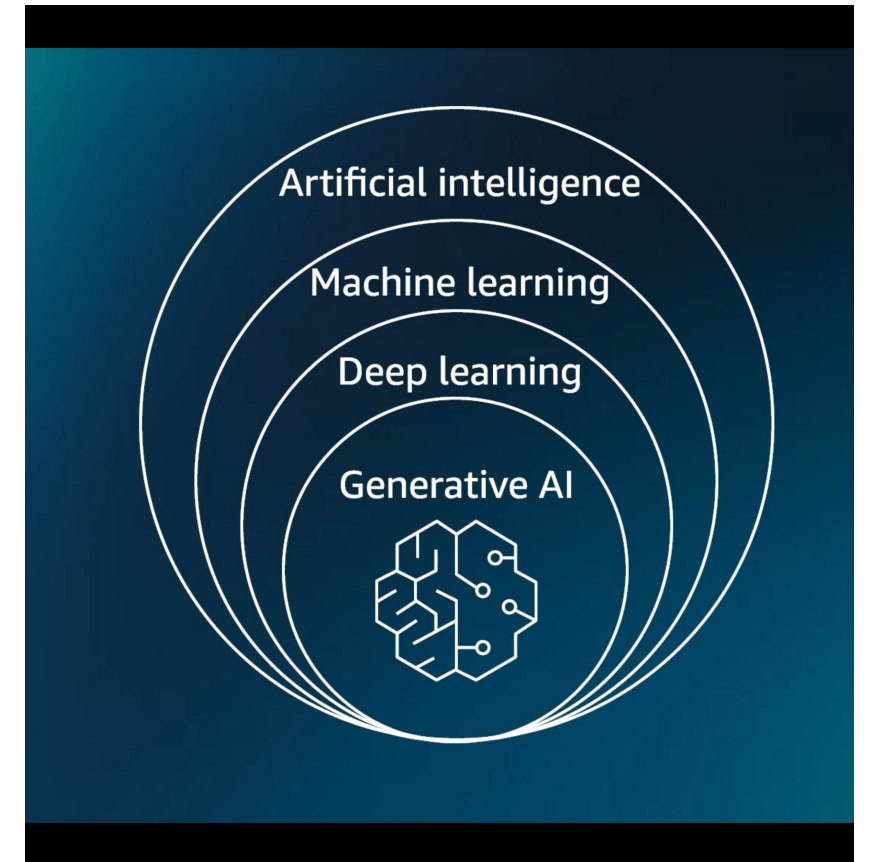


Artificial Intelligence

Dr. Ahmed Al-Jarah

Artificial Intelligence (AI)

- Artificial Intelligence is the science and engineering of creating machines that can perform tasks requiring human intelligence, such as:
 - Perception
 - Reasoning
 - Learning
 - Problem-solving
 - Decision-making
- AI serves as an umbrella term for various techniques and approaches, including machine learning, deep learning, and generative AI, among others.



AI's Approaches



Thinking
Humanly



Acting Humanly



Thinking
Rationally



Acting
Rationally



Thinking Humanly

- The exciting new effort to make computers think; machines with minds.
 - The automation of activities that we associate with human thinking; activities such as decision-making, problem-solving, learning .
-

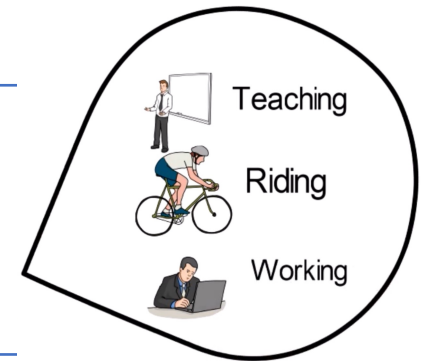
How human thinks

Through Introspection

Thought 1 : I will go.
Thought 2 : I am at rest.
Thought 3 : Its lunch time.

Through Psychological experiments

- Observing the person's act
- Humans can recognize faces, emotions, and patterns easily



Brain Imaging

- Humans can create new ideas, art, and concepts
- Not just recognize patterns, but invent new ones





Acting Humanly

- The art of creating machines that perform functions that require intelligence when performed by people.
 - The study of methods for enabling computers to perform tasks that currently require human intelligence and expertise.
-



Thinking Rationally

- The scientific study of human mental faculties through the development and analysis of computational models.
 - The study of the computational processes and mechanisms that enable perception, reasoning, learning, and action.
 - Thinking rationally refers to:
 - Designing systems that use formal logic
 - Applying rules to derive conclusions
-

How machines think rationally

- The approaches on the right evaluate performance against an ideal standard known as rationality.

5 + 5 = 6 Wrong

5 + 5 = 10 Right

- When a machine develops correct logic, it becomes capable of rational thinking.

Is Nueten mortal?

Nueten is a man

All men are mortal

Therefore, Nueten is mortal

Built an intelligent system

- The laws of thought are rules that explain how humans think clearly and correctly. Studying these rules led to the development of logic.
- In artificial intelligence, the logic-based approach tries to use these ideas to build intelligent systems.
- We must be careful about two things when designing logic systems:
 - The logic must be correct.
 - There is a difference between solving a problem in theory and solving it in real life.
- Developing logic, can design a machine with rationally thinking

Example: Navigation, Getting from A to B

In theory:

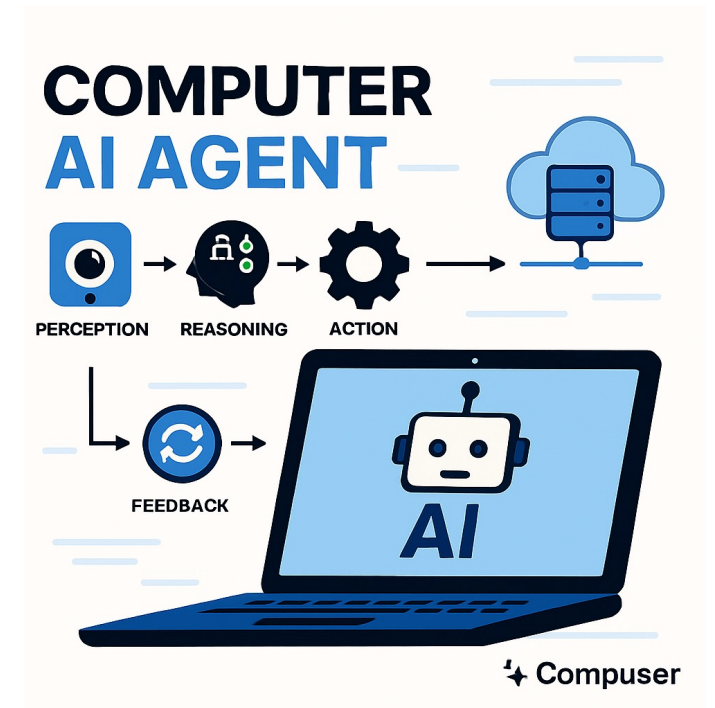
- You use Shortest Path Problem to find the fastest route.
- The map is perfect
- No traffic, accidents, or road closures
- The algorithm gives you the optimal path
- Result: You get the mathematically shortest or fastest route.

In real life (when you actually drive):

- There is traffic
- Roads may be closed
- Weather can affect conditions
- People make unpredictable decisions.
- The best route in theory might:
 - Be slower in reality
 - Become unusable suddenly

Acting rationally

- Computational Intelligence is the study of how to design intelligent agents.
- AI focuses on creating machines and systems that show intelligent behavior.
- Computer AI agents are expected to do:
 - Operate autonomously: can work on its own without constant human control.
 - Perceive the environment: can collect information from its surroundings using data or sensors.
 - Adapt to change: can adjust its behavior when the environment changes.
 - Learn from experience: ability of improve its performance over time by using past data, outcomes, or feedback.
 - Reasoning: can analyze information logically to solve problems.
 - Make decisions: based on data and reasoning, chooses the best possible action.
 - Create and achieve the goals: has objectives and takes actions to reach them.



AI agent

- An AI agent is a system that can perceive its environment, make decisions, and take actions to achieve a goal.
 - **Perceives:** collects information (data, sensors, input)
 - **Processes:** analyzes the information
 - **Acts:** takes an action based on its decision
- Example: a self-driving car:
 - Perceives: cameras detect road, cars, pedestrians
 - Process (thinks): decides speed and direction
 - Acts: steers, accelerates, brakes

Solve problems in Artificial Intelligence

To solve the problem of building an AI system you should take the following steps:

- Clearly define the problem, including all requirements and what counts as a correct solution.
- Analyze the problem carefully, since some details strongly affect how it should be solved.
- Identify and represent the background knowledge needed to solve it.
- Select the most suitable method to find the solution.

Water jug problem in AI

You are given two jugs, a 4-liter one and a 3-liter one.

Neither has any measuring markers on it.

There is a pump that can be used to fill the jugs with water.

How can you get exactly 2 liters of water into a 4-liter jug?



Water jug problem solution:

- State: (x, y)
- where x represents the quantity of water in a 4-liter jug and y represents the quantity of water in a 3-liter jug.
- That is:
 - $x = 0, 1, 2, 3, \text{ or } 4$
 - $y = 0, 1, 2, 3$
- Start state: $(0, 0)$.
- Goal state: $(2, n)$ for any n .
- Here need to start from the current state and end up in a goal state.



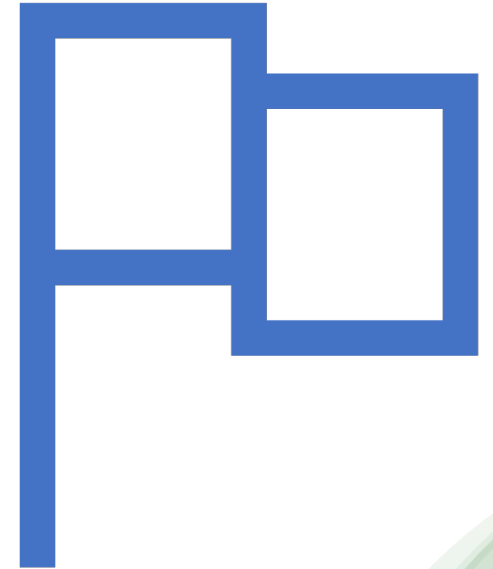
Water jug problem solution:

Production Rules for Water Jug Problem in Artificial Intelligence

1	$(x, y) \text{ is } X < 4 \rightarrow (4, Y)$	Fill the 4-liter jug
2	$(x, y) \text{ if } Y < 3 \rightarrow (x, 3)$	Fill the 3-liter jug
3	$(x, y) \text{ if } x > 0 \rightarrow (0, y)$	Empty the 4-liter jug on the ground
4	$(x, y) \text{ if } y > 0 \rightarrow (x, 0)$	Empty the 3-liter jug on the ground
5	$(x, y) \text{ if } X+Y \geq 4 \text{ and } y > 0 \rightarrow (4, y-(4-x))$	Pour water from the 3-liter jug into the 4-liter jug until the 4-liter jug is full
6	$(x, y) \text{ if } X+Y \geq 3 \text{ and } x > 0 \rightarrow (x-(3-y), 3)$	Pour water from the 4-liter jug into the 3-liter jug until the 3-liter jug is full.
7	$(x, y) \text{ if } X+Y \leq 4 \text{ and } y > 0 \rightarrow (x+y, 0)$	Pour all the water from the 3-liter jug into the 4-liter jug.
8	$(x, y) \text{ if } X+Y \leq 3 \text{ and } x > 0 \rightarrow (0, x+y)$	Pour all the water from the 4-liter jug into the 3-liter jug.
9	$(0, 2) \rightarrow (2, 0)$	Pour the 2-liter water from the 3-liter jug into the 4-liter jug.

Apply the production rules:

1. Current state = (0, 0)
2. Loop until the goal state (2, 0) reached
 - Apply a rule whose left side matches the current state
 - Set the new current state to be the resulting state
 - (0, 0) – Start State
 - (0, 3) – Rule 2, Fill the 3-liter jug
 - (3, 0) – Rule 7, Pour all the water from the 3-liter jug into the 4-liter jug.
 - (3, 3) – Rule 2, Fill the 3-liter jug
 - (4, 2) – Rule 5, Pour water from the 3-liter jug into the 4-liter jug until the 4-liter jug is full.
 - (0, 2) – Rule 3, Empty the 4-liter jug on the ground
 - (2, 0) – Rule 9, Pour all the water from the 3-liter jug into the 4-liter jug



Tic-Tac-Toe game playing in AI

- Assume ,
 - Player 1 – X
 - Player 2 – O
 - Computer could be X or O
 - Player1 –X start the game
- Player who gets 3 consecutive marks first, they will win the game.

1	2	3
4	5	6
7	8	9

2-D game-board



1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---

1-D Vector

Board and Move Table

Consider a Board having nine elements vector. Each element will contain:

0 for blank

1 indicating X player move

2 indicating O player move

0	1																												
<table><tr><td>X</td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	X									<table><tr><td>X</td><td></td><td></td></tr><tr><td></td><td>O</td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	X				O														
X																													
X																													
	O																												
2	3	4 X-win																											
<table><tr><td>X</td><td></td><td>X</td></tr><tr><td></td><td>O</td><td></td></tr><tr><td></td><td></td><td></td></tr></table>	X		X		O					<table><tr><td>X</td><td></td><td>X</td></tr><tr><td></td><td>O</td><td></td></tr><tr><td>O</td><td></td><td></td></tr></table>	X		X		O		O			<table><tr><td>X</td><td>X</td><td>X</td></tr><tr><td></td><td>O</td><td></td></tr><tr><td>O</td><td></td><td></td></tr></table>	X	X	X		O		O		
X		X																											
	O																												
X		X																											
	O																												
O																													
X	X	X																											
	O																												
O																													

Index	Current Board	New Board	Player
0	000000000	100000000	X
1	100000000	100020000	O
2	100020000	101020000	X
3	101020000	101020200	O
4	101020200	111020200	X (WIN)



Tic Tac Toe algorithm

To make a move:

1. Treat the board as a ternary number and convert it to decimal.
2. Use this number as an index to look up the move in the table.
3. Replace the current board with the resulting board state from the table.

Importance of Artificial Intelligence

- Rapid growth driven by big data, AI analyzes large datasets faster and more accurately than humans.
- Enables pattern recognition and deeper insights from data.
- Provides powerful tools and methods for business problem-solving, make better decisions and solve complex problems

In programming, AI focuses on:

- symbolic programming, work with experiences
- problem-solving techniques
- search algorithms

Activity1: The Monkey and Banana problem is a famous game problem in Artificial Intelligence

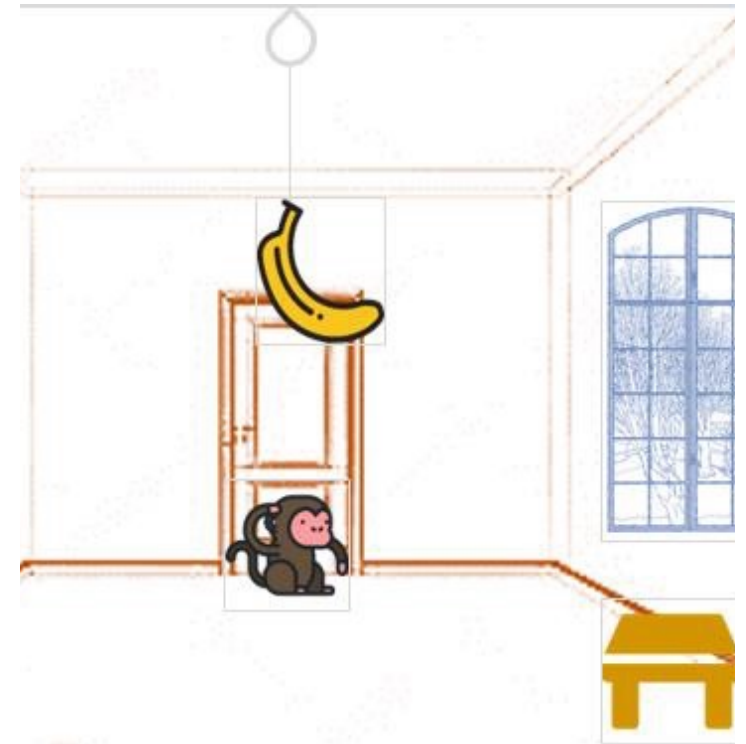
- It is mainly used in Logic programming, planning systems and problem-solving in AI

Scenario:

- A monkey is inside a room
- There is a box in the room
- A bunch of bananas is hanging from the ceiling
- The bananas are out of the monkey's reach

Goal:

- The monkey wants to get the bananas
- Idea:
- The monkey must:
 - Think
 - Plan
 - Use the box
 - Take actions in the correct order





(1)

(2)

(3)



(1)

(2)

(3)



(1)

(2)

(3)